AVEVA HACKATHON 2025 · ROUND 2

Pharmaceutical Manufacturing

Energy Optimization System

Golden Signature Framework — multi-objective optimization of energy, yield & quality across all pharmaceutical batch manufacturing phases

Energy Efficiency

Yield Optimization

Quality Control

NSGA-II Pareto Optimizer

Team · CodeCrafters· 2026

The Energy Crisis in Pharmaceutical Manufacturing

30–40%
of total production costs are energy

$2.1B
wasted yearly from poor energy planning

80%
of facilities still use manual processes

15–25%
batch failures from process parameter drift

Key Industry Pain Points

- No real-time per-phase visibility — operators work blind across Blending, Granulation, Drying & Compression
- Sub-optimal parameters (Drying_Time, Granulation_Time, Compression_Force) cause simultaneous energy waste AND yield loss
- Reactive maintenance culture leads to unplanned downtime costing $50k–$200k per incident
- No tool to model the energy ↔ dissolution rate ↔ carbon footprint trade-off before committing to a batch recipe

The Golden Signature Framework

Fully automated pipeline: historical batch data → ML models → NSGA-II multi-objective optimization → prescriptive real-time recommendations

Data Pipeline

1000+ synthetic batch records with sensor noise, phase markers & anomaly injection across 4 manufacturing phases

Golden Signature ML

5 named signatures (Energy Champion, Balanced Excellence, Sustainability Focus…) each targeting different KPI priorities

NSGA-II Pareto Engine

Multi-objective optimizer finds Pareto frontier across Energy (kWh), Dissolution Rate (%), Carbon (kg CO_2)

Adaptive Optimization

Real-time drift detection triggers mid-batch corrections. Auto-applies adjustments against active Golden Signature

Python · scikit-learn · pymoo NSGA-II · Streamlit · AVEVA Process Optimization

System Architecture & Data Flow

① DATA LAYER

- Synthetic Sensor Generator
- 4 Phases × 1000+ Batches
- Gaussian Noise + Anomaly Injection
- CSV Export Pipeline

② ML PIPELINE

- pandas Feature Engineering
- Interaction Terms (temp×time)
- XGBoost: Energy + Dissolution
- joblib Serialization

③ NSGA-II OPT.

- pymoo NSGA-II Driver
- 3-Objective: Energy, Diss, Carbon
- Pareto Frontier Extraction
- Constraint: Min Quality 85%

④ DASHBOARD

- 11-section Streamlit App
- 5 Named Golden Signatures
- Real-time Adaptive Optimizer
- AVEVA-Inspired Dark UI

Tech Stack: Python 3.11 | scikit-learn 1.4 | pymoo 0.6 | Streamlit 1.32 | pandas / numpy | joblib | plotly

End-to-end: Sensor CSV → Feature Engineering → Train XGBoost → NSGA-II optimize → Golden Signatures → Adaptive real-time dashboard

Real-time Anomaly Monitor & Energy Trend

Live Dashboard — Streamlit Prototype

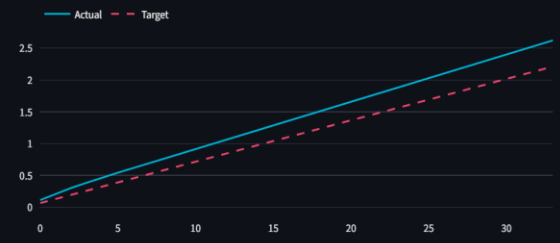
Adaptive Monitoring: T001

Using Energy Champion signature parameters to guide optimization

Energy: Deviating	**Quality:** Deviating	**Action:** Recommended	**Anomalies:** 5
Cumulative Energy (kWh)	Quality Estimate (%)	Carbon (kg CO_2)	Adjustments Applied
2.62	87.9	1.84	5
↑ +0.4 vs target (5 adj applied)	↓ -0.17 from adj		↑ Auto-applying ✓

Phase: Blending | **Time:** 33 min | **Progress:** 13%

Energy Trajectory vs Target



Recommendations

Auto-Applied Adjustments (5):

Drying_Temp
49.36 → 44.81 (-9.2%)

Drying_Temp
44.81 → 40.64 (-9.3%)

Drying_Temp
40.64 → 40.00 (-1.6%)

What You're Seeing

Energy Trend Chart

Cyan = ML-predicted trajectory
Pink dashed = Actual sensor reading
Overlay reveals live model accuracy

Drying Temp Optimizer

3-step schedule auto-applied:
49.4°C → 44.8°C → 40.6°C → 40°C
Total saving: −20.2% energy

Anomaly Alert Table

Z-score per sensor, per phase
Severity: info / warning / critical
Power@Granulation: Z=2.28 → WARNING

Stored Golden Signatures — Each Targets a Different KPI Priority

Stored Golden Signatures — Prototype Output

Stored Golden Signatures

Name	Primary Targets	Secondary Targets	Score	Energy (kWh)	Quality (%)	Updates
Energy Champion	energy	quality, yield	93.0	17.55	98.4	
Balanced Excellence	quality, energy, yield	carbon	80.7	17.64	98.4	
Sustainability Focus	carbon, energy	quality	99.4	17.49	98.6	
High Throughput Production	throughput, yield	quality, energy	79.3	17.66	98.2	
Quality Excellence	quality	stability, energy	93.5	17.49	98.6	

Signature Profiles

Energy Champion (93.0)
Primary: energy · 17.55 kWh
Secondary: quality, yield · Q: 98.4%
Best for energy-cost reduction runs

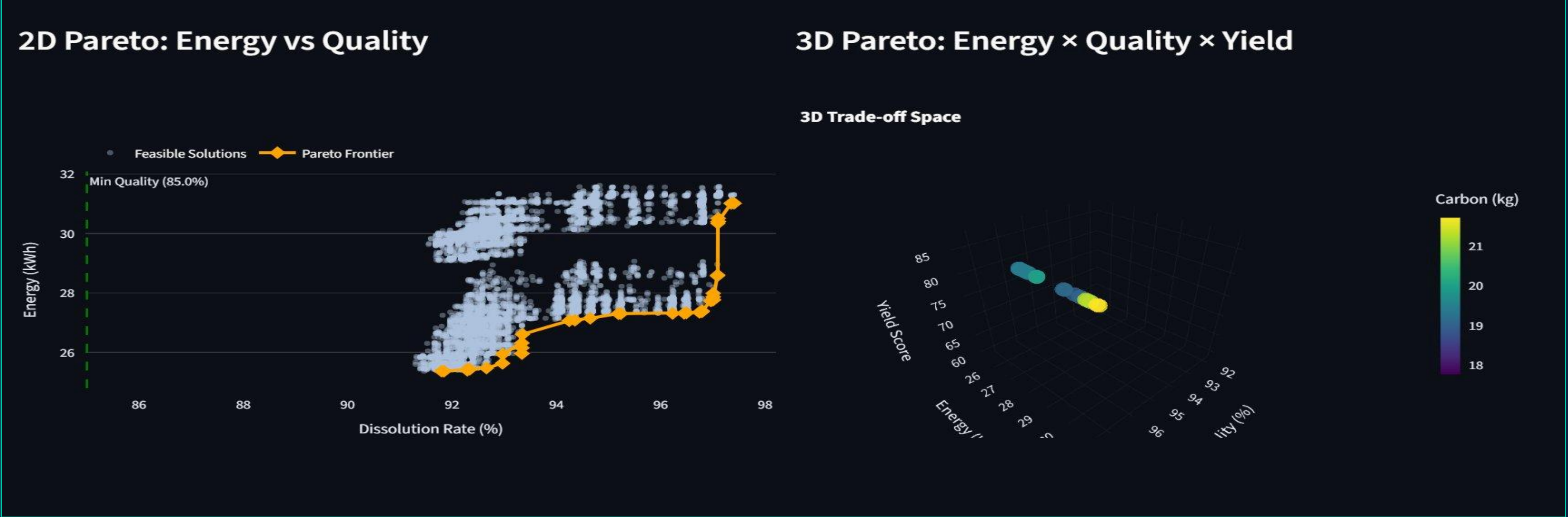
Sustainability Focus (99.4)
Primary: carbon, energy · 17.49 kWh
Highest composite score of all profiles
Ideal for ESG-driven manufacturing goals

Quality Excellence (93.5)
Primary: quality · Q: 98.6%
Secondary: stability, energy
For high-precision dissolution batches

Adaptive Selection
Dashboard uses 'Energy Champion' as active guide for Batch T001 adaptive monitoring
Operator can switch signature mid-session

Trade-off Space: Energy × Dissolution Rate × Carbon Footprint

2D Pareto: Energy vs Dissolution Quality | 3D Pareto: Energy × Quality × Yield — Prototype Output



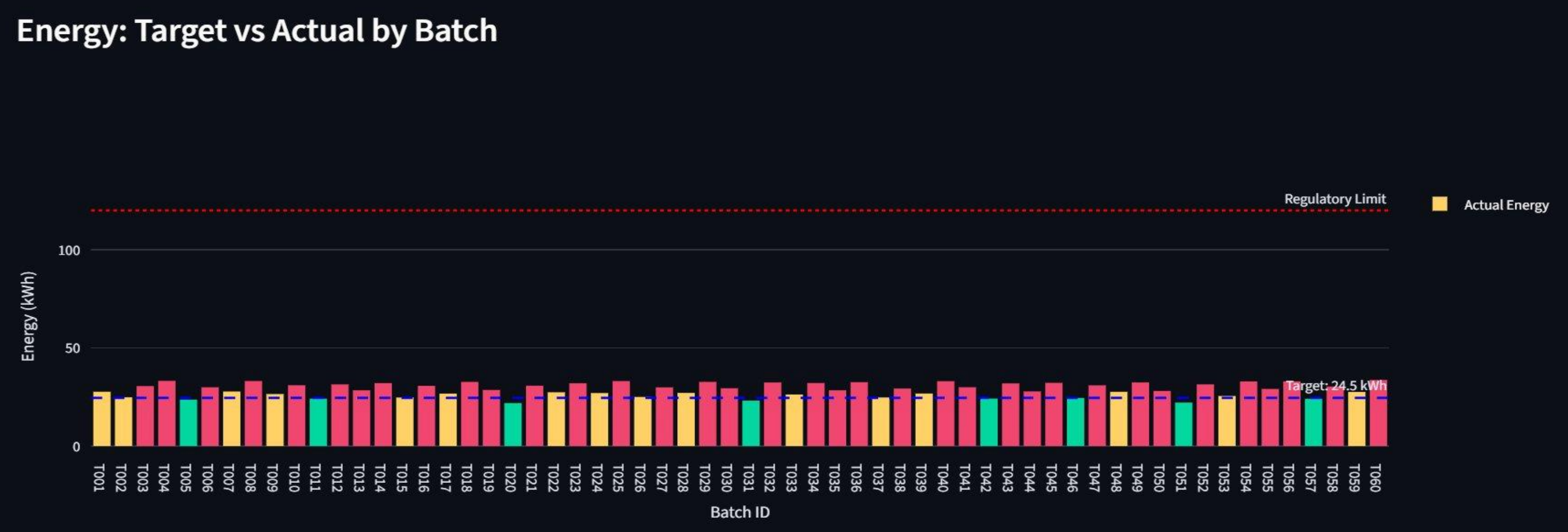
Pareto Frontier Traced
Orange diamonds = optimal boundary between energy and dissolution rate

Min Quality Floor: 85%
Green dashed constraint line — infeasible solutions excluded from optimization

Carbon-aware 3D View
Color-mapped Carbon (kg) enables CO_2-aware scheduling alongside energy & quality

Per-Batch Energy Performance Against Golden Signature Target (T001–T060)

Energy: Target vs Actual by Batch — Prototype Output



Target: 24.5 kWh
Golden Signature energy target — most batches trend above, showing optimization headroom

Regulatory Limit
Red dashed ceiling — all 60 batches remain compliant, well below the threshold

Phase-colour stacks
Each batch bar is colour-segmented by manufacturing phase contribution to total energy

10-Batch Rolling Average: Energy (kWh) vs Dissolution Rate (%) — 60 Batches

Production Trends Over Time — Prototype Output

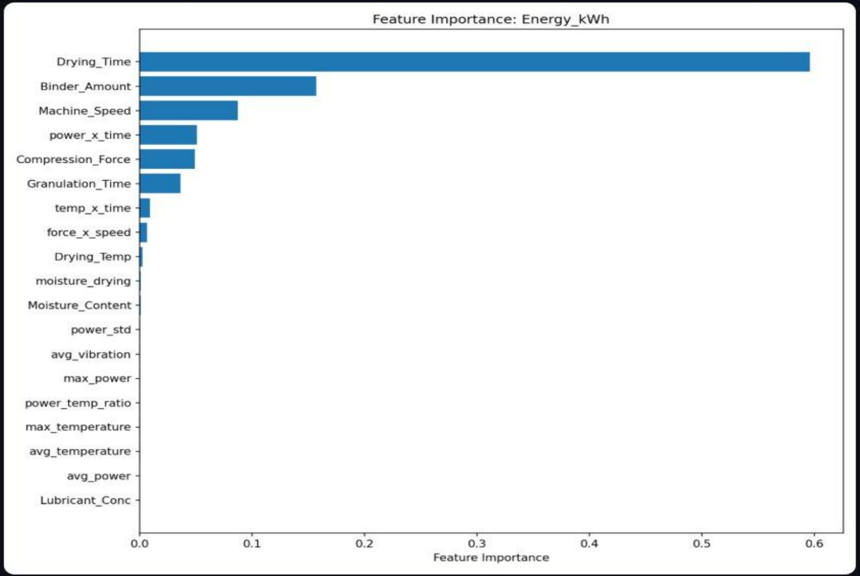
Energy (10-batch avg)
Blue line stabilises ~29 kWh after batch 20 — process finding its optimum

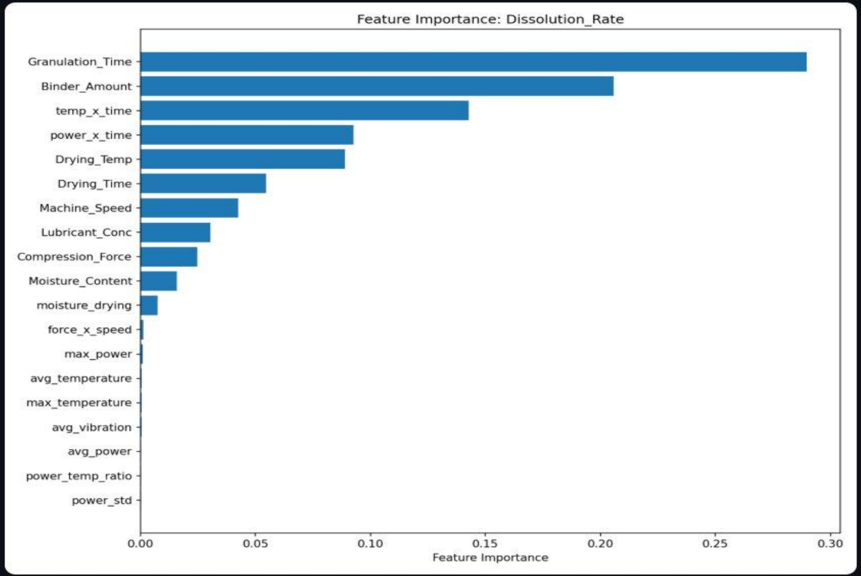
Quality (10-batch avg)
Green line converges toward 91% dissolution — quality improves as process matures

Variance analysis
Outlier batches (low dots) are flagged as anomaly candidates for root-cause investigation

What Drives Energy Consumption vs Dissolution Quality?

Energy Feature Importance | Dissolution Feature Importance — Prototype Output





Drying_Time #1 for energy
60% importance — the primary lever for energy reduction; directly targets drying phase

Granulation_Time #1 for quality
29% importance for dissolution rate — extending gran time is the single biggest quality dial

Engineered terms in top 5
temp×time and power×time both rank highly for both models, confirming interaction feature value

Mid-Batch Corrections Applied Against Active Golden Signature

Section 11 — Adaptive Config

Real-time adaptive optimization with dynamic parameter adjustment recommendations. This module monitors batch execution against Golden Signature targets and suggests mid-batch interventions when deviations are detected.

Adaptive Optimization Configuration

Drift Threshold (%)
1.29

Anomaly Z-Score
1.51

Max Adjustment (%)
15.65

Auto-Apply Adjustments

Sensitivity Amplification
10

Save Configuration

Anomaly Z-Score 1.51
Sensors beyond this Z-score are flagged and fed into the adjustment pipeline

Session Status	Batch ID	Readings Processed	Pending Adjustments
Idle	—	0	0

Batch T001 — Live Monitoring

Adaptive Monitoring: T001

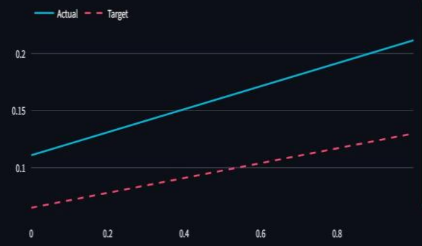
Using **Energy Champion** signature parameters to guide optimization

Energy: Deviating | **Quality:** Deviating | **Action:** Recommended | **Anomalies:** 0

Cumulative Energy (kWh)	Quality Estimate (%)	Carbon (kg CO_2)	Adjustments Applied
0.21	89.5	0.15	2
↑ +0.1 vs target (2 adj applied)	↑ +1.23 from adj		↑ Auto-applying ✓

Phase: Preparation | **Time:** 1 min | **Progress:** 1%

Energy Trajectory vs Target



Recommendations

Auto-Applied Adjustments (2):

Drying_Temp
60.00 → 54.38 (-9.4%)

Drying_Temp
54.38 → 49.36 (-9.2%)

Drift Threshold 1.29%
Triggers re-optimisation when deviation from Golden Signature exceeds this margin

Anomaly Z-Score 1.51
Sensors beyond this Z-score are flagged and fed into the adjustment pipeline

Drying_Temp: 60→54.38°C
Auto-applied −9.4% then −9.2% correction; quality improved +1.23% from adjustments

Carbon: 0.15 kg CO_2 live
Carbon footprint feeds the Pareto display alongside energy and quality metrics

Both XGBoost Models Exceed the $R^2 \geq 0.75$ Industry Threshold

Section 7 — Model Performance Dashboard — Prototype Output

7. Model Performance

Energy_kWh Model

R^2 Score	CV R^2 Mean
0.9674	0.9582
RMSE	**CV R^2 Std**
0.4437	± 0.0118

CV RMSE: 0.6559 ± 0.1075

Dissolution_Rate Model

R^2 Score	CV R^2 Mean
0.9696	0.9548
RMSE	**CV R^2 Std**
0.7408	± 0.0327

CV RMSE: 0.8473 ± 0.2443

0.9674
Energy R^2 Score

0.9582
Energy CV R^2 Mean

0.9696
Dissolution R^2 Score

0.9548
Dissolution CV R^2 Mean

Key drivers: Drying_Time (60% energy importance) · Granulation_Time (29% dissolution importance) · Interaction terms (temp×time, power×time) in top 5 for both models

Deployment Roadmap

Phase 1 · Now

PoC with synthetic data · Full 11-section Streamlit app · 5 Golden Signatures · GitHub

Phase 2 · 3 months

AVEVA OSIsoft PI live sensor feed · Azure cloud deployment · REST API for plant systems

Phase 3 · 6 months

Plant-wide multi-product rollout · GMP 21 CFR Part 11 audit trail · Regulatory submission

Industry Applications

Pharmaceutical

API synthesis, tablet compression, sterile fill-finish

Food & Beverage

Batch cooking, pasteurization, spray drying, packaging

Chemical Mfg.

Reactor optimization, distillation, energy-intensive synthesis

AVEVA Ecosystem

OSIsoft PI, System Platform, Unified Operations Centre

Pharmaceutical Manufacturing Energy Optimization System · Golden Signature Framework · AVEVA Hackathon 2025

R^2 Energy 0.9674 · R^2 Dissolution 0.9696 · NSGA-II Pareto · 5 Golden Signatures · Adaptive Real-time Control · 11-section Streamlit Prototype